NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Sports & Arthroscopy

1. ACL and PCL
2. menisci and cartilage
3. patellofemoral instability
4. rotator cuff
5. shoulder instability

ACL and PCL

1. ACL anatomy and biomechanics

- **Origin:** posteromedial aspect of the **lateral femoral condyle** (medial wall of the lateral condyle, intercondylar notch). Landmarks: **lateral intercondylar ridge ("resident's ridge")** marks the anterior (upper in flexion) limit of the footprint; **lateral bifurcate ridge** separates the AM and PL bundles.
- **Insertion:** tibia, anterior to and between the intercondylar eminences, lateral to the medial tibial spine; **anterior horn of the lateral meniscus** blends with it.
- **Length about 32-38 mm, width about 10-11 mm**. **Two bundles (named by tibial insertion):**
  - **Anteromedial (AM): tight in flexion**; primary restraint to anterior tibial translation (anterior drawer at 90 degrees).
  - **Posterolateral (PL): tight in extension**; resists **rotational** loads (pivot shift) and anterior translation near extension.
- **Blood supply: middle genicular artery** (synovial sheath); **nerve:** posterior articular branches of the **tibial nerve** (mechanoreceptors: proprioception).
- **Function: primary restraint to anterior tibial translation (about 85% at 30 and 90 degrees of flexion)**; secondary restraint to internal rotation and varus/valgus.
- **Strength:** about **2,160 N** ultimate tensile load (native).

2. ACL injury

- **Mechanism: non-contact** (about 70%): **deceleration, cutting, pivoting, landing** with the knee near extension, **valgus and external (or internal) rotation**: "**pop**", immediate swelling (**haemarthrosis: ACL tear in about 70% of acute traumatic haemarthroses** in adults).
- **Females 2-8 times higher risk: narrower notch**, **smaller ACL**, **increased Q angle/valgus**, **quadriceps dominance (less hamstring activation)**, **landing in a more upright position with knee valgus**, hormonal (preovulatory phase), **increased posterior tibial slope** (lateral). **Neuromuscular training programmes** (e.g. **FIFA 11+, PEP**) reduce ACL injury risk.

- **Examination:**
  - **Lachman test** (**most sensitive**; knee at **20-30 degrees** of flexion),
  - **Anterior drawer** (90 degrees; less sensitive acutely, hamstring guarding, posterior horn of medial meniscus acts as a "doorstop"),
  - **Pivot shift** (**most specific**; best under anaesthesia; anterior subluxation of the lateral tibial plateau in extension reducing at **20-40 degrees of flexion** via the **iliotibial band** as it moves posterior to the axis of rotation; **requires an intact IT band and MCL**),
  - **KT-1000 arthrometer** (**side-to-side difference over 3 mm** abnormal).
- **Imaging:**
  - **Radiographs: Segond fracture** (**lateral capsular avulsion of the lateral tibial plateau: anterolateral ligament (ALL)/capsule**: **pathognomonic of ACL tear**), **lateral femoral notch sign** (deep sulcus over 1.5-2 mm in the lateral femoral condyle), tibial spine avulsion (children).
  - **MRI:** discontinuity, **bone bruises ("kissing contusions") on the lateral femoral condyle (sulcus terminalis) and posterolateral tibial plateau** (pivot-shift injury), anterior tibial translation over 5-7 mm, uncovered posterior horn of the lateral meniscus, buckled PCL.
- **Associated injuries:**
  - **Meniscal tears: acute: lateral meniscus more commonly torn** (posterior horn, **root tears**); **chronic ACL deficiency: medial meniscus (posterior horn) tears** increase (it becomes the secondary restraint).
  - **Ramp lesions** (**meniscocapsular separation of the posterior horn of the medial meniscus**: probe through the **posteromedial portal/trans-notch view**; often missed on MRI).
  - **MCL injuries** (commonest associated ligament), **posterolateral corner (PLC)** (**missed PLC = commonest cause of ACL graft failure from a missed injury**), chondral injuries, **"unhappy triad" (O'Donoghue: ACL + MCL + medial meniscus; actually lateral meniscus more often in acute injuries)**.

3. Management of ACL rupture

- **Non-operative:** "copers" (low-demand, no instability, willing to modify activities): **progressive rehabilitation**. **KANON trial (Frobell):**

rehabilitation + optional delayed reconstruction gave similar 2- and 5-year outcomes to early reconstruction in young active adults (**about 50% avoided surgery**). **Increased risk of secondary meniscal tears** with recurrent instability.

- **Operative indications: recurrent instability**, **high-demand pivoting sports**, **repairable meniscal tear**, **combined ligament injuries**, **young age (children: growth-respecting techniques)**, occupational demands.
- **Timing: after resolution of swelling and recovery of full extension and quadriceps control** (avoids **arthrofibrosis**); the "**prehabilitation**" principle. **Early surgery** (within 2-3 weeks) is acceptable when ROM is good; **lock knee from a bucket-handle tear: urgent arthroscopy**.

Graft choice

Graft	Advantages	Disadvantages
Bone-patellar tendon-bone (BPTB)	**Bone-to-bone healing (6-8 weeks), lowest re-rupture in young athletes, strong (about 2,900 N)**	**Anterior knee pain, kneeling pain**, patellar fracture, patellar tendon rupture, **patella baja**, extension loss
Hamstring (semitendinosus ± gracilis, quadrupled)	**Strongest (about 4,000 N quadrupled)**, small incision, **less anterior knee pain**	**Soft-tissue healing (8-12 weeks)**, **tunnel widening**, hamstring weakness (deep flexion), saphenous nerve (infrapatellar branch) injury, **higher re-rupture in young/female athletes with small (under 8 mm) grafts**
Quadriceps tendon (± bone block)	Thick, large cross-section, **low donor morbidity**, increasing use in primary and revision	Quadriceps weakness, patellar fracture (with bone block)
Allograft	No donor morbidity, shorter surgery	**Higher failure in young (under 25), active patients** (especially **irradiated (over 2.5 Mrad)**), slower incorporation, disease transmission (rare), cost

- **Graft diameter: under 8 mm** hamstring grafts have higher revision rates.
- **Graft incorporation: avascular necrosis**, **revascularisation**, **cellular repopulation** and **remodelling (ligamentisation)**: **weakest at about 6-12 weeks**; full maturation 1-2 years.

Technical points

- **Anatomic tunnel placement**: **femoral tunnel low on the lateral wall of the notch (within the native footprint, below the lateral intercondylar ridge)**, drilled via **anteromedial portal**, **outside-in**, or flexible reamers (transtibial drilling often creates a **vertical, non-anatomic graft**: rotational instability with a persistent pivot shift).
- **Commonest technical error: femoral tunnel placed too anteriorly** (graft **tight in flexion**: loss of flexion or graft failure/stretching).
- **Tibial tunnel too anterior**: **roof impingement in extension** (loss of extension, **cyclops lesion** risk); too posterior: vertical graft, PCL impingement.
- **Fixation:** interference screws, suspensory buttons (femoral), staples/screw-post (tibial).
- **Lateral extra-articular tenodesis (LET; modified Lemaire)** or **ALL reconstruction** added in **high-risk patients** (young, high-grade pivot shift, return to pivoting sport, generalised laxity, revision): **STABILITY trial** (hamstring ACLR + LET in young patients: **reduced graft rupture/persistent rotatory laxity**).
- **Double-bundle reconstruction:** no clear clinical advantage over anatomic single-bundle.
- **ACL repair** (proximal avulsion types) is being revisited with augmentation; not standard.

Rehabilitation and return to sport

- **Early full extension, quadriceps activation, closed kinetic chain exercises** (less ACL graft strain than open chain in the early phase); **avoid open-chain quadriceps (knee extension) exercises between 0-45 degrees** early.
- **Return to sport: criteria-based** (strength over 90% limb symmetry index, hop tests, psychological readiness) and **not before 9 months**: each month

delay in return to sport until 9 months reduces re-injury by about 51% (Grindem, Delaware-Oslo).

- **Contralateral ACL injury** risk is similar to graft rupture risk in young athletes.

Complications

- **Arthrofibrosis/loss of extension** (commonest complication; **cyclops lesion** = fibroproliferative nodule anterior to the graft: **loss of terminal extension with a "clunk"**: arthroscopic excision).
- **Graft failure: early (technical: tunnel malposition, fixation failure)**, **late (re-injury)**; **missed PLC/medial injuries**; **increased posterior tibial slope (over 12 degrees)**.
- **Infection** (**coagulase-negative staphylococci**, *S. aureus*; **pre-soak graft in vancomycin** reduces infection), **tunnel widening**, DVT, saphenous nerve injury, anterior knee pain, **osteoarthritis** (higher in meniscectomy cases; reconstruction does not prevent OA).
- **Revision ACLR:** assess **tunnel position, widening (over 15 mm: staged bone grafting)**, **posterior tibial slope (over 12 degrees: slope-reducing osteotomy)**, **varus alignment (HTO)**, missed associated laxities.

4. Paediatric ACL injuries

- **Tibial spine (eminence) fractures** (**Meyers and McKeever classification**: I undisplaced, II hinged anterior elevation, III completely displaced (IIIA, IIIB rotated), IV comminuted): **types I and reducible II = closed reduction in extension, cast**; **III/IV and irreducible II = arthroscopic/open fixation (sutures or screws)**; **entrapment of the intermeniscal ligament or anterior horn of medial meniscus** blocks reduction.
- **Intrasubstance ACL tears in skeletally immature:** early reconstruction (to avoid meniscal/chondral damage) using **physeal-sparing (all-epiphyseal, Micheli-Kocher extraphyseal ITB)** or **transphyseal** techniques (with soft-tissue grafts, small vertical central tunnels, no hardware across the physis) depending on **remaining growth (Tanner stage, bone age)**; risk of **growth disturbance (angular deformity, leg-length discrepancy)**.

5. PCL anatomy and biomechanics

- **Origin: lateral surface of the medial femoral condyle** (anterior notch); **insertion: posterior tibia in a central depression (PCL facet), about 1-1.5 cm below the joint line**.
- **Twice as strong and wider than the ACL** (about 13 mm wide); **two bundles (named by femoral origin): anterolateral (AL, larger, tight in flexion)** and **posteromedial (PM, tight in extension)**.
- **Meniscofemoral ligaments** (from the posterior horn of the lateral meniscus to the medial femoral condyle): **Humphrey (anterior to the PCL)** and **Wrisberg (posterior to the PCL)** (at least one in about 90% of knees).
- **Primary restraint to posterior tibial translation (about 90-95%)**; secondary restraint to external rotation (with PLC).

6. PCL injury

- **Mechanism: dashboard injury** (posteriorly directed force on the proximal tibia with the knee flexed), **fall onto a flexed knee with the foot plantarflexed**, **hyperextension** (combined injuries).
- **Examination:**
  - **Posterior drawer test** (**most sensitive**; 90 degrees of flexion): **grading by tibial step-off** relative to the femoral condyles (**normal about 1 cm anterior step-off**): **grade I: 0-5 mm translation (tibia still anterior to condyles)**, **grade II: 5-10 mm (tibia flush with the condyles)**, **grade III: over 10 mm (tibia posterior to the condyles: suggests combined PCL + PLC/posteromedial injury)**.
  - **Posterior sag (Godfrey) sign**, **quadriceps active test** (contraction reduces posterior subluxation), **reverse pivot shift** (PLC).
  - **Dial test** (external rotation asymmetry over 10 degrees): **at 30 degrees only = isolated PLC**, **at 30 and 90 degrees = PLC + PCL**.
- **Imaging: stress radiographs** (posterior displacement **over 12 mm** side-to-side suggests combined injury), MRI, **avulsion fractures of the tibial PCL insertion**.

Management

- **Isolated grade I-II PCL injuries: non-operative**: **quadriceps-strengthening rehabilitation**, **PCL brace (dynamic anterior drawer force, e.g. Jack brace)**: good functional outcomes; natural history may include **medial compartment and patellofemoral OA** in the long term.
- **Bony avulsions of the PCL tibial insertion: early fixation** (posterior approach (Burks and Schaffer) or arthroscopic) gives excellent results.
- **Grade III, combined injuries (PLC/multiligament), symptomatic chronic instability: reconstruction**: **transtibial** (**"killer turn"** of the graft at the posterior tibia: graft abrasion/attenuation) vs **tibial inlay** (open posterior bone block; avoids the killer turn); **single- (AL bundle)** or **double-bundle**.
- **Risks: popliteal artery injury** during tibial tunnel drilling (**guide-wire penetration; keep the knee flexed at 90 degrees**, use a guard, posteromedial portal to visualise).

7. Posterolateral corner (PLC)

- **Components: lateral (fibular) collateral ligament (LCL)**, **popliteus tendon**, **popliteofibular ligament** (the "big three"), plus lateral capsule, arcuate ligament, biceps femoris, IT band, lateral gastrocnemius.
- **Function:** primary restraint to **varus** and **external rotation** (especially at 30 degrees).
- **Injury:** posterolateral force to the anteromedial tibia, hyperextension/varus; **common peroneal nerve injury in about 15-30%**.
- **Signs: varus thrust gait**, **dial test**, **external rotation recurvatum test**, **posterolateral drawer**, **reverse pivot shift**, **varus stress at 30 degrees (and at 0 degrees: suggests combined cruciate injury)**.
- **Imaging: arcuate sign** (**avulsion of the fibular styloid**: attachments of the popliteofibular, arcuate and fabellofibular ligaments): PLC injury; varus stress radiographs.
- **Treatment: acute repair within about 2-3 weeks** (higher failure of repair alone) or **reconstruction (anatomic, e.g. LaPrade)**: **reconstruction more reliable than repair**; address **varus alignment (HTO first)** in chronic cases; **PLC must be addressed with any cruciate reconstruction** (unrecognised PLC injury causes cruciate graft failure).

8. Multiligament knee injury/knee dislocation

- **Schenck classification: KD I** single cruciate + collateral; **KD II** both cruciates; **KD III M/L** both cruciates + **MCL (M) or PLC (L)**; **KD IV** both cruciates + both collaterals; **KD V** fracture-dislocation; C = arterial injury, N = nerve injury.
- **Vascular injury (popliteal artery)** in about **10-30%** (anterior dislocations: intimal stretch; posterior: transection). **ABI under 0.9: CT angiography/vascular surgery**; **normal pulses do not exclude injury**: serial **ankle-brachial index (ABI)** ± angiography. Ischaemia over 6-8 hours: high amputation rate; **prophylactic fasciotomy** after revascularisation.
- **Peroneal nerve injury** in about 25% (especially **KD III L**).
- **Management:** reduce urgently, assess vascular status, **external fixation** for unstable/vascular/open injuries; **early (within 3 weeks) single-stage or staged ligament repair/reconstruction** generally gives better results than non-operative treatment; **arthrofibrosis** is common.

Exam pearl: **Lachman most sensitive; pivot shift most specific (needs intact ITB/MCL). Segond fracture = ACL tear. Acute ACL: lateral meniscus tears; chronic: medial meniscus. Commonest technical error: femoral tunnel too anterior (tight in flexion). Tibial tunnel too anterior: roof impingement, cyclops lesion.**

Exam pearl: **AM bundle tight in flexion, PL bundle tight in extension (ACL named by tibial insertion); PCL AL bundle tight in flexion (named by femoral origin). Humphrey anterior, Wrisberg posterior to the PCL.**

Exam pearl: **PCL grade III (tibia behind condyles) = suspect PLC. Dial test: 30 only = PLC; 30 and 90 = PLC + PCL. Arcuate sign = fibular styloid avulsion. Missed PLC causes cruciate graft failure. Knee dislocation: check ABI; under 0.9 = angiography.**

Menisci and cartilage

1. Meniscal anatomy

- **Fibrocartilage**: about **70% water**; dry weight mostly **type I collagen (about 90%)**; proteoglycans, cells (**fibrochondrocytes**).
- **Collagen orientation: circumferential fibres (deep, dominant)** convert axial load into **hoop stresses**; **radial ("tie") fibres** resist longitudinal splitting; superficial random mesh.
- **Medial meniscus: C-shaped**, wider posteriorly, **firmly attached to the capsule and deep MCL**: **less mobile (about 2-5 mm excursion)**: **torn more often** (overall). **Lateral meniscus: more circular (O-shaped)**, covers more of the plateau, **more mobile (about 9-11 mm)**; attached to the **popliteus** (popliteal hiatus: no capsular attachment there) and **meniscofemoral ligaments (Humphrey, Wrisberg)**.
- **Blood supply: medial, lateral and middle genicular arteries** via a **perimeniscal capillary plexus**; only the **peripheral 10-30% (medial) and 10-25% (lateral)** is vascular in adults.
  - **Red-red zone** (peripheral, **under 3 mm from the rim**): good healing.
  - **Red-white zone** (**3-5 mm**): moderate healing.
  - **White-white zone** (**over 5 mm**, inner): avascular, poor healing.
- **Function: load transmission** (**about 50% of load in extension and about 85-90% in flexion through the menisci**; **lateral meniscus bears about 70% of lateral compartment load** vs medial about 50%), **shock absorption**, **joint stability** (**posterior horn of the medial meniscus is a secondary restraint to anterior tibial translation**, especially in ACL-deficient knees), **lubrication and nutrition**, proprioception.
- **Total meniscectomy increases contact stresses by about 2-3 times (up to 235%)**: **Fairbank changes** on radiographs after meniscectomy: **ridge (osteophyte) formation, flattening of the femoral condyle, joint space narrowing**. **Lateral meniscectomy** has worse outcomes than medial (lateral compartment convex-on-convex geometry).

2. Meniscal tears

- **Patterns:**

  - **Longitudinal (vertical): bucket-handle** (displaced longitudinal: **locked knee**; **double PCL sign** on MRI, **"fragment in the notch" sign**; commoner medially),
  - **Radial** (transects circumferential fibres: **loss of hoop stress function**; **"ghost meniscus"** sign),
  - **Horizontal (cleavage)** (degenerative; **parameniscal cysts**, especially lateral),
  - **Oblique (flap/parrot-beak)**, **complex/degenerative**,
  - **Root tears** (within about 1 cm of the root attachment): **functionally equivalent to total meniscectomy**; **medial posterior root tears in middle-aged women** with a "pop" on squatting (**meniscal extrusion over 3 mm** on MRI); **lateral posterior root tears with ACL injuries**,
  - **Ramp lesions** (posterior medial meniscocapsular junction; ACL-associated),
  - **Meniscal extrusion** (over 3 mm: root tear or degeneration).
- **Clinical:** joint-line tenderness (**most sensitive sign**), **McMurray test**, **Thessaly test (at 20 degrees)**, **Apley grind**; locking, effusion (delayed, hours).
- **MRI: sensitivity about 90-95%** (medial better than lateral); **grade 3 signal (intrasubstance signal reaching the articular surface)** = tear; grades 1-2 are intrasubstance degeneration (not tears).

3. Management of meniscal tears

- **Degenerative tears in middle-aged/older knees** (with or without OA): **physiotherapy first**; **arthroscopic partial meniscectomy (APM) is no better than sham surgery or physiotherapy** for degenerative tears (**FIDELITY**, **ESCAPE**, Katz **MeTeOR**); APM is reserved for **true mechanical symptoms (locking)** or failure of conservative care.
- **Repair indications ("save the meniscus"): peripheral (red-red/red-white), longitudinal, vertical tears 1-4 cm**, **young patient**, **acute**, **stable knee or with concurrent ACL reconstruction** (**healing rates higher with ACL reconstruction: bleeding, growth factors**), **root tears**, **ramp lesions**, **bucket-handle tears** in young patients (reduce and repair).
- **Repair techniques:**

  - **Inside-out** (**gold standard** for mid-body/posterior tears; vertical mattress sutures; **posteromedial approach (protect the saphenous nerve)**; **posterolateral approach: protect the peroneal nerve (retract biceps femoris posteriorly, work anterior to the lateral gastrocnemius)**),
  - **Outside-in** (anterior horn tears),
  - **All-inside** (posterior horn; device-based; **risk to popliteal neurovascular structures in the lateral posterior horn**),
  - **Vertical mattress sutures are biomechanically strongest** (capture circumferential fibres).
  - **Biological augmentation:** marrow venting (notch microfracture), **fibrin clot**, rasping/trephination.
- **Root repair: transtibial pull-out** repair or suture anchor; slows extrusion and OA progression vs partial meniscectomy (in knees without advanced OA).
- **Rehabilitation after repair:** protected weight bearing/limited deep flexion (**avoid squatting beyond 90 degrees for about 3-4 months**).
- **Outcomes: repair success about 70-90%** (better laterally, with ACL reconstruction, acute, peripheral); **failure rate higher medially and in isolated repairs**.
- **APM outcomes:** good for symptomatic unstable flaps in younger knees, but **long-term OA risk** proportional to the amount removed.

4. Discoid meniscus

- **Lateral (almost always)**; **about 3-5% of the population (more in Asians, up to about 15%)**; **bilateral in about 20%**.
- **Watanabe classification:**
  - **Type I: complete** (covers the entire plateau),
  - **Type II: incomplete** (covers under 80%),
  - **Type III: Wrisberg variant** (**no posterior meniscotibial (coronary) attachment**, only the **meniscofemoral ligament of Wrisberg**: **hypermobile**: **"snapping knee syndrome"** in children).
- **Imaging:** radiographs: **widened lateral joint space**, squared lateral condyle, cupping of the lateral plateau; **MRI: three or more contiguous 5-**

mm sagittal slices showing a continuous "bow-tie" (meniscal body continuity), or **transverse diameter over 15 mm or over 20% of tibial width**.

- **Treatment: asymptomatic: observe**; **symptomatic: arthroscopic saucerisation (partial meniscectomy to a rim of about 6-8 mm)** ± **repair of peripheral instability (Wrisberg type)**; **avoid total meniscectomy** (early lateral OA, osteochondritis dissecans of the lateral femoral condyle).

5. Meniscal allograft transplantation (MAT) and scaffolds

- **Indication: young (under about 50), symptomatic post-meniscectomy knee** (pain in the meniscus-deficient compartment) **with minimal arthritis (Outerbridge under III)**, **normal alignment (or corrected by osteotomy)**, **stable knee (or concomitant ACL reconstruction)**, **BMI under about 35**.
- **Contraindications:** advanced OA, uncorrected malalignment/instability, inflammatory arthritis, obesity.
- **Size-matched fresh-frozen allograft** (radiographic sizing: **Pollard method**); fixation with **bone plugs (medial) or a bone bridge/keyhole (lateral: horns close together)**, or soft-tissue sutures.
- **Outcomes: about 70-80% survival at 10 years**; better laterally; **shrinkage, extrusion, tears** common.
- **Collagen (CMI) and polyurethane (Actifit) scaffolds** for partial segmental defects (intact rim needed).

6. Articular cartilage: structure

- **Hyaline cartilage: about 65-80% water**; **type II collagen (about 90-95% of collagen; tensile strength)**; **proteoglycans (aggrecan: GAG chains of chondroitin and keratan sulphate bound to hyaluronic acid by link protein)**: **negative charge attracts water: compressive stiffness (swelling pressure)**; **chondrocytes (about 1-5% of volume)**: produce matrix; **avascular, aneural, alymphatic**: **nutrition by diffusion from synovial fluid**: poor healing.
- **Other collagens: type IX** (cross-links type II), **type X (calcified zone, hypertrophic chondrocytes)**, **type XI** (regulates fibril size), **type VI** (pericellular).
- **Zones:**

  - **Superficial (tangential):** collagen **parallel** to the surface (**highest collagen, resists shear**), **lowest proteoglycan**, flattened cells, **lamina splendens**, **highest water content**,
  - **Middle (transitional):** oblique collagen, round cells,
  - **Deep (radial):** collagen **perpendicular** to the surface, **highest proteoglycan, lowest water**, columns of cells (resists compression),
  - **Tidemark** (separates uncalcified from calcified cartilage),
  - **Calcified cartilage** (type X collagen; anchors to subchondral bone).
- **Lubrication: boundary lubrication (lubricin: proteoglycan 4)**, **fluid-film**, **weeping/elastohydrodynamic**; **biphasic** (solid + fluid) viscoelastic material.
- **Ageing vs OA:**

Parameter	Ageing	Osteoarthritis
Water content	**Decreases**	**Increases (early)**
Collagen	Unchanged/cross-linking increases	Network disrupted
Proteoglycan content	Decreases	**Decreases**
Chondroitin sulphate	**Decreases**	**Increases (chondroitin 4-sulphate)**
Keratan sulphate	**Increases**	**Decreases**
Modulus of elasticity	**Increases (stiffer)**	**Decreases**
Degradative enzymes (**MMP-1, -13, ADAMTS-4/5**, IL-1, TNF)	Normal	**Increased**
Chondrocyte number	Decreases; size increases	Early proliferation (clusters/cloning)

7. Cartilage injury and repair

- **Partial-thickness injuries (not crossing the tidemark): do not heal** (no access to marrow cells).
- **Full-thickness (through the subchondral plate):** marrow cells form **fibrocartilage (type I collagen)**: mechanically inferior to hyaline.

- **Classification: Outerbridge** (grade I softening; **II fissuring/fragmentation under 1.5 cm (0.5 inch) diameter**; **III over 1.5 cm**; **IV exposed subchondral bone**), **ICRS** (0 normal; 1 superficial; 2 **under 50% depth**; 3 **over 50% depth (to calcified layer)**; 4 to subchondral bone).
- **Treatment algorithm for focal chondral defects** (young patients, **correct alignment, stability and meniscal deficiency first**):

Defect	Low-demand	High-demand
Small (under 2 cm^2)	**Debridement/chondroplasty**, **microfracture/marrow stimulation**	**Osteochondral autograft transfer (OATS/mosaicplasty)**, microfracture
Large (over 2-4 cm^2)	Microfracture (less durable), **ACI/MACI**	**Autologous chondrocyte implantation (ACI/MACI)**, **osteochondral allograft (OCA)** (especially with bone loss)

- **Microfracture (Steadman): debride to a stable vertical rim**, **remove the calcified cartilage layer**, awls **3-4 mm apart, 2-4 mm deep**; **fibrocartilage**; **protected weight bearing and CPM for 6-8 weeks**; **results deteriorate after about 2-5 years**, best in **young (under 40), small (under 2-4 cm^2), femoral condyle** lesions; **intralesional osteophytes (subchondral bone overgrowth)** complicate later ACI.
- **OATS (osteochondral autograft): hyaline cartilage** transfer from **less weight-bearing areas** (**lateral trochlea, intercondylar notch**); **defects under about 2 cm^2**; **donor site morbidity** (patellofemoral pain); plugs must be **flush** (**proud plugs: high contact pressures**; recessed: fibrocartilage overgrowth).
- **ACI/MACI: two-stage** (**arthroscopic biopsy of cartilage**, culture, **implantation at about 4-6 weeks** under a **periosteal patch (first generation: hypertrophy commonest complication)**, **collagen membrane (second)** or **matrix-seeded (MACI, third generation)**); **hyaline-like** repair tissue; **larger defects (2-10 cm^2)**; better than microfracture for larger defects in some trials; **slow rehabilitation (about 12 months)**.
- **Osteochondral allograft (fresh, refrigerated, used within about 28 days)**: **large defects with subchondral bone loss**, **OCD**, AVN, salvage;

viable chondrocytes (over 70% viability required); **no immune rejection of cartilage (chondrocytes protected in matrix)**, bone portion incorporates via **creeping substitution**.

- **Subchondral insufficiency fracture of the knee (SIFK, formerly SONK): elderly women**, **medial femoral condyle**, often after a **medial meniscus posterior root tear**; **sudden pain**; MRI: **subchondral low-signal line with bone marrow oedema**; protected weight bearing; UKA/TKA if collapse.

8. Osteochondritis dissecans (OCD)

- **Subchondral bone lesion** (repetitive microtrauma, vascular, genetic) with **risk of instability of the overlying cartilage**; **lateral aspect of the medial femoral condyle (about 70%)** (**Wilson sign**: pain at 30 degrees of flexion with internal rotation, relieved by external rotation).
- **Juvenile OCD (open physes)** has a **much better prognosis** (**about 50-75% heal with non-operative care**: activity restriction 3-6 months) than **adult OCD** (closed physes: usually need surgery).
- **MRI signs of instability (De Smet/Kijowski): high T2 signal line behind the fragment**, **cysts** beneath the lesion, **articular fracture line**, **fluid-filled osteochondral defect**.
- **Treatment:**
  - **Stable lesions in juveniles: activity modification, immobilisation**; **retrograde/antegrade drilling** if not healing after 3-6 months.
  - **Unstable but salvageable: fixation** (**bioabsorbable or metal compression screws, chondral darts**, bone pegs).
  - **Unsalvageable/loose bodies: excision + cartilage restoration (OATS, OCA, ACI)**.
- **Other sites: capitellum** (**throwers/gymnasts**: see elbow notes), **talar dome** (**posteromedial (deeper, cup-shaped, atraumatic)** and **anterolateral (shallow, wafer-shaped, traumatic)**).

Exam pearl: **Hoop stresses via circumferential fibres; radial and root tears abolish them (root tear = total meniscectomy). Red-red zone (under 3 mm) heals; repair with ACL reconstruction heals best; vertical mattress strongest. Degenerative tears: physio first (FIDELITY, ESCAPE).**

Exam pearl: **Discoid lateral meniscus: Watanabe III (Wrisberg) is hypermobile: saucerise + repair. MAT: young, meniscectomised, minimal OA, aligned, stable.**

Exam pearl: **Cartilage: type II collagen, aggrecan; superficial zone parallel collagen/highest water; deep zone highest proteoglycan. Ageing: water down, stiffer, keratan up; OA: water UP, keratan down, chondroitin up, MMP-13/ADAMTS up. Microfracture: fibrocartilage (type I); OATS: hyaline, small; ACI/MACI: larger defects; OCA: large osteochondral defects.**

Exam pearl: **OCD: lateral aspect of the medial femoral condyle; juvenile heals non-operatively more often; high T2 line behind the fragment = unstable. SIFK: elderly women, medial femoral condyle, medial root tear.**

Patellofemoral instability

1. Anatomy and biomechanics

- **Patella:** the **largest sesamoid**; **thickest articular cartilage in the body (up to about 5-7 mm)**; **lateral facet larger** than the medial; **odd facet** (medial, contacts in deep flexion).
- **Function: increases the moment arm of the quadriceps (by about 30-50%)**, centralises forces, protects the knee.
- **Contact areas:** at **20 degrees**, the **distal pole** contacts the trochlea; contact moves **proximally** on the patella with increasing flexion; **maximum contact area at about 90 degrees**; **at about 120-135 degrees**, the odd facet contacts the medial femoral condyle.
- **Patellofemoral joint reaction force: about 0.5 x body weight walking**, **3-4 x stairs**, **7-8 x squatting**; increases with flexion (and with open-chain extension near full extension: high quadriceps force).
- **Engagement:** the patella is **laterally positioned in extension** and **engages the trochlear groove at about 20-30 degrees of flexion**: **instability occurs in early flexion (0-30 degrees)**, where the **soft-tissue restraints matter most**.

Restraints to lateral patellar translation

- **Medial patellofemoral ligament (MPFL): primary restraint (about 50-60%) at 0-30 degrees**. **Femoral attachment (Schöttle point): between the adductor tubercle and the medial epicondyle** (just anterior to the posterior cortex extension line, proximal to the posterior origin of the medial condyle on a true lateral radiograph); **patellar attachment: proximal two-thirds of the medial border**. **Injured in almost all (about 90-100%) first-time dislocations**, commonly at the **femoral attachment** (adults) or **patellar attachment** (children).
- **Medial patellotibial ligament, medial patellomeniscal ligament, medial retinaculum** (secondary, more important in deeper flexion), **vastus medialis obliquus (VMO)** (dynamic).
- **Trochlear geometry**: the main restraint **beyond about 30 degrees** of flexion.

2. Risk factors for instability (Dejour "major instability factors")

1. **Trochlear dysplasia** (**most important; about 85-96% of patients with recurrent instability**).
2. **Patella alta** (**delays engagement of the patella in the trochlea**).
3. **Increased tibial tubercle-trochlear groove (TT-TG) distance** (**over 20 mm abnormal**; normal about 10-13 mm; lateralised tubercle).
4. **Patellar tilt** (over 20 degrees on CT).
5. **Other factors: generalised ligamentous laxity (Beighton score)**, **femoral anteversion** (increased), **external tibial torsion**, **genu valgum (increased Q angle)**, **VMO dysplasia**, **female sex**, **young age (adolescents)**, **previous dislocation**, family history.
6. **Q angle:** line from ASIS to the patellar centre and from the patellar centre to the tibial tubercle: **normal about 14 degrees (men) and about 17 degrees (women)**; measured in extension (and in 30 degrees flexion to seat the patella).

3. Imaging and measurements

Radiographs

- **True lateral (posterior condyles superimposed):**
  - **Crossing sign** (**trochlear floor line crosses the anterior border of the lateral condyle**: **trochlear dysplasia**: the trochlea becomes flat at that point),
  - **Supratrochlear spur (bump)**,
  - **Double contour** (hypoplastic medial facet).
  - **Patellar height indices:**
  - **Insall-Salvati ratio** (**patellar tendon length / patellar length**; **normal 0.8-1.2**; **over 1.2 = alta**, under 0.8 = baja),
  - **Caton-Deschamps index** (distance from the **inferior pole of the patellar articular surface to the anterosuperior tibial plateau / articular surface length**; **normal 0.6-1.2**; **over 1.2 = alta**; **not affected by tibial tubercle osteotomy**: preferred for planning distalisation),
  - **Blackburne-Peel ratio** (perpendicular distance from the tibial plateau line to the inferior articular pole / articular length; **over 1.0 = alta**).

- **Axial (Merchant view, 45 degrees knee flexion): sulcus angle (normal about 138 ± 6 degrees; over 145 = dysplasia)**, **congruence angle (normal -6 ± 11 degrees; over +16 abnormal: lateral subluxation)**, **lateral patellofemoral angle** (Laurin: should open laterally).

CT/MRI

- **TT-TG distance** (**over 20 mm = abnormal**; MRI values are smaller than CT by about 2-4 mm), **TT-PCL distance (over 24 mm)** (less affected by trochlear dysplasia/knee rotation).
- **Trochlear dysplasia (Dejour classification):**
  - **Type A: crossing sign**; shallow trochlea (sulcus angle over 145 degrees) but preserved morphology,
  - **Type B: crossing sign + supratrochlear spur**; flat or convex trochlea,
  - **Type C: crossing sign + double contour**; asymmetric facets (convex lateral, **hypoplastic medial facet**),
  - **Type D: crossing sign + spur + double contour**; **"cliff" pattern** (vertical step between facets).
- **Patellar tilt**, **femoral anteversion** and **tibial torsion** (rotational profile CT), **MRI**: **MPFL tear**, **bone bruise** (**medial patellar facet and lateral femoral condyle**: pathognomonic of transient dislocation), **osteochondral fractures/loose bodies**.

4. Acute first-time patellar dislocation

- **Mechanism: twisting on a flexed, valgus knee with the foot planted** (internal rotation of the femur on a fixed tibia), or direct blow; usually **lateral** dislocation; often reduces spontaneously with knee extension.
- **Examination:** haemarthrosis, **tenderness over the medial retinaculum/MPFL femoral attachment (Bassett sign)**, **apprehension sign (Fairbank)**, **patellar glide (quadrants: over 2-3 quadrants lateral translation = laxity)**.
- **Osteochondral fractures** in about **25-50%** (**inferomedial patellar facet** (commonest) or **lateral femoral condyle**): radiographs/MRI.
- **Management:**
  - **Non-operative for most first-time dislocations** without loose bodies: **brief immobilisation for comfort**, **early ROM, quadriceps (VMO) and**

hip abductor/external rotator strengthening, **patellar stabilising brace/taping**.

- **Surgery after the first dislocation if: osteochondral fracture/loose body** (**fix or remove: arthroscopy**), **substantial MPFL disruption with lateral patellar subluxation on the axial view (relative)**, **high-risk patients** (young with **dysplasia + patella alta + open physes: recurrence risk over 50-70%**).
- **Recurrence after the first dislocation: about 15-45% overall** (**up to about 70%** in skeletally immature with dysplasia).

5. Recurrent patellar instability: surgical options

- **Principle: address the predominant anatomical abnormalities ("à la carte" surgery, Dejour).**

MPFL reconstruction

- **Workhorse procedure** for recurrent instability: **gracilis or semitendinosus autograft**; femoral fixation at the **anatomic Schöttle point** (**fluoroscopic confirmation**); **graft fixed at about 30-60 degrees of flexion** with the patella centred in the trochlea; **avoid over-tensioning** (**medial patellofemoral overload, pain, loss of flexion, medial instability**).
- **Femoral tunnel malposition: too proximal** (graft **tight in flexion**: **loss of flexion and medial patellar overload**), **too distal** (tight in extension/loose in flexion).
- **Paediatric: femoral attachment is just distal to the distal femoral physis**: drill below the physis or use **physeal-sparing (soft-tissue/adductor sling) techniques**.
- **Complications: patellar fracture** (transverse tunnels across the patella: use **anchors or short sockets**), recurrent instability, stiffness, overconstraint.
- **MPFL repair** has higher recurrence than reconstruction.

Tibial tubercle osteotomy (TTO)

- **Medialisation (Elmslie-Trillat)**: **increased TT-TG (over 20 mm)**: **aim for TT-TG about 10-15 mm**; **avoid over-medialisation** (increases medial tibiofemoral and **medial patellofemoral** contact pressures).

- **Anteromedialisation (Fulkerson)**: **oblique osteotomy**: medialises + **anteriorises (unloads the distal and lateral patellar facets)**: **lateral facet or distal patellar chondral lesions** with instability/malalignment; **contraindicated for proximal/medial facet lesions** (shifts load onto them).
- **Distalisation**: **patella alta (Caton-Deschamps over about 1.2-1.3)**: target about 1.0.
- **Complications: proximal tibial fracture** (**avoid early full weight bearing with Fulkerson: risk of tibial fracture especially at 6-12 weeks**), non-union, **compartment syndrome**, **DVT**, **anterior tibial artery/deep peroneal nerve injury (posterior cortex drilling)**, painful hardware, **Hauser procedure (posteromedialisation: increases PFJ pressure: abandoned; causes PFJ arthritis)**.
- **Roux-Goldthwait** (lateral half of the patellar tendon transferred medially) in skeletally immature (avoids the tibial tubercle apophysis: **TTO contraindicated with open physes: genu recurvatum**).

Trochleoplasty

- **Severe trochlear dysplasia (Dejour B and D with a supratrochlear spur over 5 mm)**, especially **revision/failed MPFL reconstruction**: **sulcus-deepening trochleoplasty (Dejour/Bereiter)**, **recession wedge trochleoplasty** (for spur). **Complications:** arthrofibrosis, cartilage necrosis, PFJ arthritis; **contraindicated with open physes and established PFJ arthritis**.

Lateral retinacular release

- **Never as an isolated procedure for instability** (no benefit, can cause **iatrogenic medial patellar instability**). **Lateral lengthening** for a **tight lateral retinaculum with a positive patellar tilt test (tilt under 0 degrees: cannot lift lateral facet to neutral)**.

Rotational/angular osteotomies

- **Femoral derotational osteotomy** for **excessive femoral anteversion (over about 25-30 degrees)**; **distal femoral varus osteotomy** for **genu valgum** (over about 5 degrees mechanical valgus) in recurrent instability; **tibial derotation** for external tibial torsion.

6. Special situations

- **Habitual (obligatory) dislocation:** dislocates with **every flexion**, in children: **quadriceps contracture** (**vastus lateralis/iliotibial band contracture**; prior quadriceps injections): **quadricepsplasty/lengthening with realignment**.
- **Congenital (permanent) dislocation:** irreducible from birth; associated with **Down syndrome, nail-patella syndrome (LMX1B: hypoplastic patella, iliac horns, renal disease)**, **arthrogryposis**: early soft-tissue realignment.
- **Medial patellar instability: iatrogenic after lateral release** (vastus lateralis tendon detachment); **lateral retinaculum reconstruction/repair**.
- **Patellofemoral pain syndrome (anterior knee pain): diagnosis of exclusion**; **hip abductor/external rotator and quadriceps strengthening** (best evidence), taping, orthoses; **no surgery**.
- **Patellar tendinopathy (jumper's knee): eccentric/heavy slow resistance exercise**, **decline squats**; **Blazina staging**; surgery (debridement) for refractory cases.
- **Sinding-Larsen-Johansson** (inferior pole traction apophysitis) and **Osgood-Schlatter** (tibial tubercle traction apophysitis): self-limiting; **excise the ossicle** after maturity if symptomatic.

7. Summary: planning algorithm

Pathology	Threshold	Procedure
Torn/insufficient MPFL	Recurrent instability	**MPFL reconstruction (all cases)**
TT-TG increased	**Over 20 mm**	**Medialising TTO** (± anteriorisation for lateral/distal facet wear)
Patella alta	**Caton-Deschamps over 1.2**	**Distalising TTO** (or MPFL alone if mild)
Severe trochlear dysplasia	**Dejour B/D, spur over 5 mm**	**Trochleoplasty** (revision mainly)
Femoral anteversion	**Over about 25-30 degrees**	**Femoral derotation osteotomy**

Pathology	Threshold	Procedure
Valgus	Mechanical valgus over about 5 degrees	**Distal femoral varus osteotomy**
Tight lateral retinaculum	Tilt test negative	**Lateral lengthening** (never isolated release)

Exam pearl: **MPFL = primary restraint 0-30 degrees; femoral attachment (Schöttle point) between the adductor tubercle and medial epicondyle; torn in nearly all dislocations. Dejour major factors: trochlear dysplasia (crossing sign), patella alta, TT-TG over 20 mm, tilt over 20 degrees.**

Exam pearl: **Bone bruise on the medial patellar facet and lateral femoral condyle = transient patellar dislocation. Osteochondral fracture of the inferomedial patella: fix or remove. First-time dislocation without loose body: non-operative.**

Exam pearl: **Proximal femoral tunnel in MPFLR = tight in flexion, medial overload. Fulkerson AMZ unloads lateral/distal facets (avoid for proximal/medial lesions). Caton-Deschamps index is used for distalisation. Never isolated lateral release (iatrogenic medial instability). TTO contraindicated with open physes (recurvatum): Roux-Goldthwait.**

Rotator cuff

1. Anatomy

Muscle	Nerve (root)	Insertion	Action	Test
Supraspinatus	**Suprascapular (C5-6)**	Greater tuberosity (superior facet)	**Abduction initiation**, depresses/compresses the head	**Jobe (empty can) test**, drop arm test
Infraspinatus	**Suprascapular (C5-6)**	Greater tuberosity (middle facet)	**External rotation (with arm at side)**	**External rotation lag sign**, resisted ER
Teres minor	**Axillary (C5-6)**	Greater tuberosity (inferior facet)	**External rotation in abduction**	**Hornblower's sign** (ER in 90 degrees abduction), **drop sign**
Subscapularis	**Upper and lower subscapular (C5-6)**	**Lesser tuberosity**	**Internal rotation**, anterior stability	**Lift-off (Gerber)**, **belly-press (Napoleon)**, **bear-hug** (most sensitive for upper subscapularis), **internal rotation lag sign**

- **Rotator cable** (crescent-shaped thickening perpendicular to the cuff fibres, from the biceps to the infraspinatus insertion): **"suspension bridge"** load transfer: **crescent tears** (within the cable) can remain functional.
- **Footprint:** supraspinatus about **12-25 mm** medial-lateral; **"critical zone" (Codman)**: **hypovascular area about 1 cm proximal to the supraspinatus insertion** (controversial).

- **Force couples: coronal plane: deltoid vs inferior cuff (infraspinatus, teres minor, subscapularis)**; **transverse plane: subscapularis vs infraspinatus/teres minor**. Tears that **disrupt a force couple** (e.g. **posterosuperior tears extending into the infraspinatus and teres minor**) cause **pseudoparalysis**.
- **Long head of biceps (LHB):** from the **supraglenoid tubercle and superior labrum**; runs in the **bicipital groove** stabilised by the **biceps pulley (SGHL + CHL)** and **transverse humeral ligament**; **subscapularis tears are associated with medial LHB subluxation**.

2. Pathogenesis

- **Intrinsic (degenerative): age** (prevalence of asymptomatic full-thickness tears rises with age: **about 25% over 60 and about 50% over 80**), **hypovascularity**, **smoking**, **diabetes**, **hypercholesterolaemia**, genetic.
- **Extrinsic: subacromial impingement (Neer)**: **acromial morphology (Bigliani: type I flat, II curved (commonest), III hooked: associated with tears)**, **critical shoulder angle (CSA over 35 degrees: cuff tears; under 30 degrees: OA)**, **os acromiale**, **acromial spurs**; **internal impingement** (**throwers: posterosuperior glenoid contacts the undersurface of the posterior supraspinatus/anterior infraspinatus in abduction-external rotation**: **articular-sided partial tears + posterosuperior labral lesions**, associated with **GIRD**).
- **Traumatic tears** (younger patients, falls, dislocations in over 40s).

3. Classification

- **Partial-thickness tears: Ellman**: **A (articular), B (bursal), C (interstitial)**; **grade 1 under 3 mm, grade 2 3-6 mm, grade 3 over 6 mm (over 50% of tendon thickness, about 12 mm)**. **PASTA lesion** = **partial articular supraspinatus tendon avulsion** (commonest partial pattern).
- **Full-thickness tear size (DeOrio and Cofield): small under 1 cm, medium 1-3 cm, large 3-5 cm, massive over 5 cm**. **Gerber definition of massive tear: complete tear of two or more tendons**.
- **Tear shape (Burkhart/Ellman): crescent** (mobile medial-lateral: direct repair to bone), **U-shaped** (**margin convergence (side-to-side) first**, then repair to bone), **L-shaped**, **massive contracted immobile**.

- **Patte classification of retraction: stage 1** close to the insertion, **stage 2** at the humeral head, **stage 3** at the glenoid.
- **Goutallier classification of fatty infiltration (CT, adapted to MRI by Fuchs): 0 normal; 1 some fatty streaks; 2 more muscle than fat; 3 equal fat and muscle; 4 more fat than muscle**. **Grade 3-4 (over 50% fat): poor outcome, high re-tear, irreversible**.
- **Tangent sign** (supraspinatus does not cross a line drawn between the superior borders of the coracoid and the scapular spine on sagittal oblique MRI): **atrophy**.
- **Hamada classification** (cuff tear arthropathy; see shoulder arthroplasty).
- **Subscapularis tears (Lafosse):** I partial upper third; II complete upper third; III complete upper two-thirds; IV complete with head centred; V complete with eccentric head (subluxation).

4. Clinical evaluation

- **Pain** (night pain, lateral arm/deltoid region), **weakness**, **painful arc (60-120 degrees)**.
- **Impingement tests: Neer sign** (passive forward flexion with internal rotation) and **Neer test** (pain relieved by subacromial lidocaine injection), **Hawkins-Kennedy test** (forward flexion 90 degrees + internal rotation).
- **Strength tests:** see table; **external rotation lag sign** (infraspinatus), **hornblower (teres minor)**, **drop arm (Codman)**.
- **Pseudoparalysis**: **active forward elevation under 90 degrees with full passive range** (massive tears).
- **Imaging:**
  - **Radiographs: acromiohumeral interval (AHI under 7 mm suggests a massive cuff tear; under 5 mm = irreparable)**, **greater tuberosity sclerosis/cysts**, **acromial morphology (outlet view)**, **critical shoulder angle**, cuff tear arthropathy changes.
  - **Ultrasound** (dynamic, cheap; operator-dependent; similar accuracy to MRI for full-thickness tears).
  - **MRI** (tear size, retraction, **atrophy/fatty infiltration**, LHB, labrum); **MR arthrography** (articular-sided partial tears).

5. Management

Subacromial pain (impingement) without full-thickness tear

- **Physiotherapy (scapular and cuff strengthening), NSAIDs, subacromial steroid injection.**
- **Subacromial decompression (acromioplasty): no better than placebo/arthroscopy only (CSAW, FIMPACT trials)**: limited role.

Partial-thickness tears

- **Non-operative first**; **under 50%**: **debridement ± acromioplasty**; **over 50% thickness (Ellman 3)** or bursal-sided over 25-30%: **repair** (**transtendon repair for PASTA** or **completion and repair**).

Full-thickness tears

- **Acute traumatic tears in active patients: early repair (within about 3-6 weeks)** (before retraction and fatty infiltration).
- **Chronic degenerative tears: trial of physiotherapy** (successful in **about 75%** in some studies), **repair if persistent symptoms**, especially **younger, active patients**, with **reparable tears (good muscle quality: Goutallier under 3)**.
- **Natural history of untreated full-thickness tears: about 50% enlarge** (over years; **symptomatic tears, larger tears and those involving the anterior cable are more likely to progress**).

Repair techniques

- **Arthroscopic** (standard) vs **mini-open** vs **open** (similar outcomes).
- **Single-row** vs **double-row** (**double-row: better footprint restoration, higher biomechanical strength, lower re-tear rates for larger tears**, but similar clinical outcomes overall) vs **transosseous-equivalent (suture-bridge)** (**most widely used for medium-large tears**).
- **Margin convergence** for U-shaped tears (reduces strain on the repair).
- **Biceps tenotomy or tenodesis** if the LHB is diseased/unstable (**tenotomy: Popeye deformity, cramping**; **tenodesis: younger/cosmetic concern**).
- **Rehabilitation: sling for about 4-6 weeks** (passive ROM early or delayed: **similar outcomes; early motion may increase re-tear in large tears**),

active at about 6 weeks, strengthening at about **12 weeks**; tendon-bone healing about **12-16 weeks**.

Outcomes and re-tears

- **Re-tear rates: about 10-20% for small/medium tears, up to 40-90% for massive tears** (MRI); **re-tears often still have good pain relief**, though **healed repairs have better strength**.
- **Risk factors for re-tear: age over 65**, **large tear size**, **retraction**, **fatty infiltration (Goutallier 2+)**, **smoking**, **diabetes**, osteoporosis, **delayed repair**, poor tendon quality, **workers' compensation**.
- **Re-tear pattern: Cho type 1 (at the footprint: commonest with single-row)** and **type 2 (at the musculotendinous junction, medial to the medial row: double-row)**.

6. Massive irreparable cuff tears

- **Options (age, pseudoparalysis, arthritis, subscapularis and teres minor status):**
  - **Non-operative**: deltoid rehabilitation (anterior deltoid programme).
  - **Arthroscopic debridement ± biceps tenotomy** (pain relief in low-demand elderly; **preserve the coracoacromial arch**: **avoid acromioplasty/CA ligament release** to prevent **anterosuperior escape**).
  - **Partial repair** (restore force couples: repair infraspinatus/subscapularis).
  - **Superior capsular reconstruction (SCR)** (**Mihata**: **fascia lata autograft (thick, 6-8 mm)** or **dermal allograft** from the superior glenoid to the greater tuberosity: restores **superior stability**; **young patients without arthritis, intact/repairable subscapularis**).
  - **Subacromial balloon spacer** (InSpace: **no benefit over debridement in START:REACTS trial**).
  - **Tendon transfers:**
  - **Posterosuperior tears (loss of ER): latissimus dorsi transfer (Gerber)** (**contraindicated with subscapularis deficiency**), **lower trapezius transfer (Elhassan; with Achilles allograft)** (more in-phase with ER).
  - **Anterosuperior/subscapularis tears: pectoralis major transfer** (**sternal head passed under the clavicular head (behind the conjoint**

tendon: more anatomic vector)), **latissimus dorsi anterior transfer**.
- **Reverse total shoulder arthroplasty** (**elderly, pseudoparalysis, cuff tear arthropathy**).
- **Interposition grafts/patch augmentation**.

7. Related conditions

- **Calcific tendinitis: supraspinatus (commonest)**, **women aged 30-50**, **hydroxyapatite (calcium hydroxyapatite) deposition**; **not degenerative** (cell-mediated). **Uhthoff stages: precalcific (fibrocartilaginous metaplasia)**, **calcific (formative phase, resting phase, resorptive phase: most painful, acute; deposit fluffy/ill-defined)**, **postcalcific (reconstitution)**. **Gärtner classification** (type I well-defined dense; II mixed; III cloudy translucent: resorptive). **Treatment: NSAIDs, physio**, **ultrasound-guided barbotage (lavage)** (best evidence), **extracorporeal shock wave therapy (ESWT)**, **arthroscopic excision**.
- **Frozen shoulder (adhesive capsulitis): diabetes (strongest association; about 10-20% of diabetics; worse prognosis)**, thyroid disease, Dupuytren, Parkinson, post-surgery, **women aged 40-60**. **Loss of active and passive ER (with arm at side) is the hallmark**; normal radiographs. **Phases: freezing (painful), frozen (stiff), thawing** (**about 1-3 years**). **Pathology: fibroblast/myofibroblast proliferation** (**Dupuytren-like**, type III collagen) in the **rotator interval and CHL**. **Treatment: intra-articular steroid injection (best early evidence)**, physiotherapy, **hydrodilatation**, **manipulation under anaesthesia (MUA)**, **arthroscopic capsular release** (**UK FROST trial**: all three (early structured physio + steroid, MUA, ACR) improved similarly; ACR more costly/risky).
- **SLAP lesions (superior labrum anterior to posterior) (Snyder): I** degenerative fraying (debride); **II superior labrum and biceps anchor detached** (**commonest**: repair in young overhead athletes; **biceps tenodesis in over 35-40 years**); **III bucket-handle tear of the labrum, biceps intact** (excise); **IV** bucket-handle extending into the biceps (repair/tenodesis depending on biceps involvement). **Tests: O'Brien (active compression)**, **Speed's**, **dynamic labral shear (O'Driscoll)**. **Spinoglenoid paralabral cyst** (suprascapular nerve compression: infraspinatus weakness).

- **Subscapularis tears and LHB instability**: **medial subluxation of the LHB** suggests a subscapularis tear.
- **Os acromiale: failure of fusion of acromial ossification centres (meso-acromion commonest)**; **fuse by about 25 years**; can cause impingement; **ORIF with bone graft** if symptomatic.

Exam pearl: **Supraspinatus/infraspinatus = suprascapular nerve; teres minor = axillary; subscapularis = upper/lower subscapular. Belly-press/bear-hug: subscapularis; hornblower: teres minor; ER lag: infraspinatus.**

Exam pearl: **Goutallier 3-4 (fat equal/greater than muscle) = irreversible, high re-tear. AHI under 7 mm = massive tear. CSA over 35 degrees = cuff tear. Acromioplasty no better than placebo (CSAW). Re-tear risk: age, size, retraction, fatty infiltration, smoking.**

Exam pearl: **Massive irreparable tears: SCR (young, no OA), LD or lower trapezius transfer for posterosuperior; pectoralis major transfer for subscapularis; RTSA for elderly pseudoparalysis/CTA. Preserve the CA arch (anterosuperior escape). Frozen shoulder: diabetes; loss of passive ER; FROST trial. SLAP II commonest (repair young; tenodesis older).**

Shoulder instability

1. Anatomy and stabilisers

- **Glenohumeral joint:** the most mobile and most commonly dislocated large joint. **Glenoid surface area is about one-third to one-quarter of the humeral head** ("golf ball on a tee"); glenoid **retroverted about 5-7 degrees**, superiorly tilted about 5 degrees; humeral head retroverted about 30 degrees.
- **Static stabilisers:**
  - **Labrum** (deepens the glenoid by about **50%**, increases contact area, anchors the IGHL; **sublabral foramen** and **Buford complex (absent anterosuperior labrum + cord-like MGHL)** are **normal variants**: do not repair).
  - **Glenohumeral ligaments:**
  - **SGHL:** restraint to **inferior translation in adduction** (with the coracohumeral ligament; **rotator interval**),
  - **MGHL: anterior translation at about 45 degrees abduction** with external rotation (absent in about 30%),
  - **IGHL complex (anterior band, axillary pouch, posterior band): the most important**: **anterior band** resists **anterior translation in 90 degrees abduction and external rotation** (the **apprehension position**); **posterior band** resists **posterior translation in flexion and internal rotation**.
  - **Coracohumeral ligament**: inferior translation in adduction, external rotation (tight in frozen shoulder).
  - **Negative intra-articular pressure**, **concavity-compression**, **adhesion-cohesion**.
- **Dynamic stabilisers: rotator cuff (concavity-compression: most important dynamic)**, **long head of biceps**, **scapular stabilisers** (scapulothoracic rhythm about **2:1 glenohumeral:scapulothoracic**).
- **Rotator interval:** triangle between **supraspinatus and subscapularis** (base: coracoid): contains **SGHL, CHL, long head of biceps**: **interval closure** reduces inferior/posterior laxity (MDI).

2. Classification of instability

- **TUBS** (**T**raumatic, **U**nidirectional, **B**ankart lesion, **S**urgery) vs **AMBRI** (**A**traumatic, **M**ultidirectional, **B**ilateral, **R**ehabilitation, **I**nferior capsular shift if surgery). **Stanmore triangle (polar types):**
  - **Polar I:** traumatic structural,
  - **Polar II:** atraumatic structural (capsular laxity/dysplasia),
  - **Polar III: muscle patterning (abnormal muscle recruitment)**, non-structural: **physiotherapy/biofeedback; avoid surgery**.
- **Direction: anterior (about 95%)**, posterior (**about 2-5%: seizures, electrocution, posterior force with adduction/internal rotation**), inferior (**luxatio erecta**: arm fixed overhead; **axillary nerve and artery injuries**), multidirectional.

3. Traumatic anterior dislocation

- **Mechanism: abduction + external rotation + extension force**.
- **Associated lesions:**
  - **Bankart lesion** (**anteroinferior labrum + IGHL avulsion from the glenoid**): **in about 90% of young patients**. **Bony Bankart** (glenoid rim fracture).
  - **Hill-Sachs lesion** (**posterosuperolateral humeral head impaction** on the glenoid rim: **in about 80-90%**).
  - **ALPSA** (**anterior labroligamentous periosteal sleeve avulsion**: labrum displaced **medially and inferiorly** on the scapular neck: heals in a malposition: **higher recurrence**; mobilise before repair).
  - **Perthes lesion** (labrum avulsed with **intact periosteum**: non-displaced).
  - **GLAD** (**glenolabral articular disruption**: labral tear + articular cartilage injury; usually stable).
  - **HAGL** (**humeral avulsion of the glenohumeral ligament**: **"J sign"** on MRI (IGHL becomes J-shaped); **recurrent instability if missed**; **open or arthroscopic repair**).
  - **Greater tuberosity fracture** (**commoner in over 40s**; reduces with the shoulder in most cases; **displacement over 5 mm (over 3 mm in overhead athletes): fix**).

  - **Rotator cuff tears** (**over 40 years: about 30%; over 60 years: about 80%**: **"terrible triad" of the shoulder**: dislocation + cuff tear + nerve injury). **Persistent weakness after reduction in an older patient = cuff tear (until proved otherwise)**: ultrasound/MRI; **early repair**.
  - **Axillary nerve injury** (**about 5-35%, mostly neurapraxia**; **older patients**): check **regimental badge sensation** and deltoid **before and after reduction**.
  - **Axillary artery injury** (elderly with atherosclerosis).
- **Reduction techniques: Kocher** (traction, external rotation, adduction, internal rotation: **risk of fracture/nerve injury**), **Hippocratic**, **Stimson** (prone, hanging weight), **Milch** (abduction + external rotation), **FARES**, **Cunningham (massage technique)**, **scapular manipulation**; **post-reduction radiographs (AP + axillary/Velpeau)**.
- **Recurrence (the most important determinant is age at first dislocation):**
  - **Under 20 years: about 70-90%** (up to about **90%** in **contact athletes**, male),
  - **20-40 years: about 50-60%**,
  - **Over 40 years: about 10-15%** (cuff tears, stiffness more common than recurrence).
- **Immobilisation in external rotation** (to approximate the Bankart lesion) has **not consistently reduced recurrence** vs internal rotation sling.
- **Primary arthroscopic stabilisation after first dislocation** in **young (under about 25) contact/collision athletes** reduces recurrence (**about 80% recurrence non-operatively vs about 10-15% after surgery**); offered selectively.

4. Bone loss: the key to surgical choice

Glenoid bone loss

- **Inverted pear glenoid** (arthroscopic appearance: loss of anteroinferior bone, **over about 25% loss**).
- **Measurement: 3D CT (en face view of the glenoid)**: **best-fit inferior circle method**, **Pico method**, **glenoid width loss as percentage**.

- **Critical bone loss: historically about 20-25%**; **"subcritical" bone loss of about 13.5-17%** still raises recurrence and worsens outcomes after arthroscopic Bankart repair (**Shaha**).

Humeral bone loss and the glenoid track

- **Glenoid track concept (Yamamoto, Di Giacomo):** the zone of contact of the glenoid on the humeral head in **abduction/external rotation**: **glenoid track width = 0.83 x D (intact glenoid width) - d (glenoid bone loss width)**.
  - **Hill-Sachs interval (HSI) = width of the Hill-Sachs lesion + bone bridge** (to the cuff insertion).
  - **HSI over the glenoid track = "off-track" (engaging) Hill-Sachs** (it can **lever over the anterior glenoid rim** in the functional range): **high recurrence after isolated Bankart repair**.
  - **HSI under the glenoid track = "on-track"**.
- **Engaging Hill-Sachs (Burkhart):** lesion aligned parallel to the anterior glenoid in a functional position (abduction/external rotation): engages.

Treatment algorithm (Di Giacomo et al.)

Glenoid bone loss	Hill-Sachs	Treatment
Under 25% (or under subcritical threshold)	**On-track**	**Arthroscopic Bankart repair**
Under 25%	**Off-track**	**Arthroscopic Bankart + remplissage**
Over 25%	On-track	**Latarjet (bone block)**
Over 25%	Off-track	**Latarjet ± humeral procedure (remplissage/bone graft)** depending on track after Latarjet

- **ISIS score (Instability Severity Index Score; Balg and Boileau): over 6 points = high recurrence after arthroscopic Bankart (about 70%): Latarjet**:
  - **Age under 20 at surgery: 2 points**,

- **Competitive sport: 2 points**,
- **Contact or forced overhead sport: 1 point**,
- **Shoulder hyperlaxity (external rotation over 85 degrees or Gagey hyperabduction over 90 degrees): 1 point**,
- **Hill-Sachs visible on AP radiograph in external rotation: 2 points**,
- **Loss of glenoid contour on AP radiograph: 2 points**.

5. Surgical procedures

Arthroscopic Bankart repair

- **Suture anchors (at least 3 anchors, at least 4 mm apart, placed on the articular margin of the glenoid face (not medially on the neck))**, **inferior capsular shift** (plication), **anchors at the 3-5 o'clock (right shoulder)** region; **beach chair or lateral decubitus**.
- **Recurrence about 5-15% (higher with bone loss, young contact athletes, ISIS over 6, fewer than 3 anchors, ALPSA)**.
- **Open Bankart** (subscapularis split/takedown): historically the gold standard; **contact athletes** in some centres; **external rotation loss**.

Remplissage ("to fill")

- **Arthroscopic capsulotenodesis of the infraspinatus tendon (and posterior capsule) into the Hill-Sachs defect**: converts the intra-articular defect to extra-articular. **Small loss of external rotation**; reduces recurrence for off-track lesions.

Latarjet (coracoid transfer)

- **Coracoid process (about 2-3 cm) with the conjoint tendon** transferred to the **anteroinferior glenoid (flush with the glenoid face, 3-5 o'clock, below the equator)** through a **subscapularis split**, fixed with **two screws**.
- **"Triple blocking effect" (Patte):**
  - **Bony effect** (extends the glenoid arc),
  - **Sling effect of the conjoint tendon** (**most important in abduction-external rotation**: dynamic sling on the lower subscapularis and capsule),

    - **Capsular repair** (to the coracoacromial ligament stump: **Bankart effect**).
- **Bristow** = tip of the coracoid only, placed vertically, fixed with one screw (less bone).
- **Complications: graft malposition (too lateral: OA; too medial: recurrence)**, **graft non-union/osteolysis/fracture**, **screw problems**, **musculocutaneous nerve injury** (enters the coracobrachialis about **5 cm below the coracoid**: beware traction), **axillary nerve injury**, **loss of external rotation**, **glenohumeral OA (long-term)**, recurrence (low, about 2-5%), infection, haematoma.
- **Free bone block (iliac crest (Eden-Hybinette), distal tibial allograft)** for **massive bone loss or failed Latarjet**.

Other

- **HAGL repair**, **humeral head allograft/disimpaction** for large Hill-Sachs, **humeral rotational osteotomy (historical: Weber)**.
- **Thermal capsulorrhaphy: abandoned** (capsular necrosis, **chondrolysis**, axillary nerve injury). **Intra-articular bupivacaine pain pumps**: **glenohumeral chondrolysis**.

6. Posterior instability

- **Traumatic posterior dislocation: seizures, electrocution, ECT**: **often missed (about 50-80% missed initially)**; **arm locked in internal rotation**, **cannot externally rotate**; **AP radiograph: "light bulb sign"**, loss of the half-moon overlap, **rim sign (over 6 mm)**; **axillary view/CT essential**. **Reverse Hill-Sachs lesion (McLaughlin lesion)** of the **anteromedial humeral head**.
    - **Treatment by defect size: under 25%**: closed reduction, immobilise in external rotation; **25-50%**: **McLaughlin procedure (subscapularis tendon transfer into the defect)** or **modified McLaughlin (lesser tuberosity transfer: Neer)**, **allograft reconstruction**; **over 50%** or chronic over 6 months: **hemiarthroplasty/arthroplasty**.
- **Recurrent posterior subluxation: athletes (football linemen, weightlifters, swimmers)**: **reverse Bankart lesion**, **Kim lesion** (incomplete, concealed avulsion of the posteroinferior labrum); **Jerk test,**

Kim test; **posterior labral repair/capsular shift**; **posterior bone block** for glenoid bone loss; **glenoid opening-wedge osteotomy** for **excess glenoid retroversion** (over about 15-20 degrees).

7. Multidirectional instability (MDI)

- **Inferior + anterior and/or posterior instability**; **generalised hyperlaxity (Beighton score)**, connective tissue disorders (**Ehlers-Danlos**), **patulous inferior capsule**, enlarged rotator interval; often **bilateral, atraumatic**.
- **Sulcus sign (over 2 cm, persistent in external rotation = rotator interval incompetence)**, **Gagey hyperabduction test** (**over 105 degrees = IGHL laxity**).
- **Treatment: prolonged rehabilitation (at least 6 months)**: rotator cuff and scapular stabiliser strengthening (**most succeed**); **surgery** for failure: **open inferior capsular shift (Neer)** or **arthroscopic capsular plication ± rotator interval closure**. **Avoid surgery in voluntary/muscle-patterning instability (Polar III)** and **EDS** (high failure).

8. Examination tests

- **Anterior: apprehension test** (abduction-external rotation), **relocation test (Jobe: posterior force relieves apprehension)**, **release (surprise) test** (**most specific**), **anterior drawer**, **load and shift**.
- **Posterior: Jerk test**, **Kim test**, posterior drawer, posterior load and shift.
- **Inferior/MDI: sulcus sign**, **Gagey**.
- **Laxity ≠ instability:** instability = symptomatic pathological translation.

> Exam pearl: **Age at first dislocation is the strongest predictor of recurrence (under 20: up to 90%). Over 40: think cuff tear and axillary nerve. IGHL anterior band = main restraint in abduction-external rotation.**

Exam pearl: **Glenoid track = 0.83D - d; HSI over the track = off-track (engaging) = add remplissage. Glenoid loss over 20-25% (subcritical 13.5-17%) = Latarjet. ISIS over 6 = Latarjet. Latarjet: sling effect of the conjoint tendon is the main mechanism; musculocutaneous nerve at risk.**

Exam pearl: **Posterior dislocation: seizure/electrocution, locked internal rotation, light bulb sign, reverse Hill-Sachs; McLaughlin transfer for 25-50% defects. HAGL = J sign. ALPSA heals medially (higher recurrence). MDI: rehab first; inferior capsular shift; avoid surgery in muscle patterning.**